

How to Build a Rhythm Game in Godot

GodotCon Boston 2026





GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>

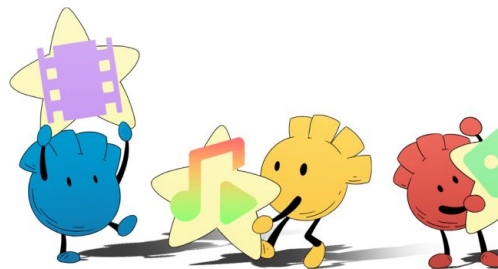


Who am I?

- Games and tools developer
- Worked with Unity, Unreal and Godot
- Built a lot of what I work on as open source projects
- Currently building my own game engine (in SDL)
- And a multi-engine tool for building rhythm games (Unity, Unreal, Godot, MonoGame, SDL, etc)
- A cat dad (most important).



GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Outline of Workshop

1. Intro to Rhythm Game Utilities plugin
2. Intro to song charting
3. Workshop

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



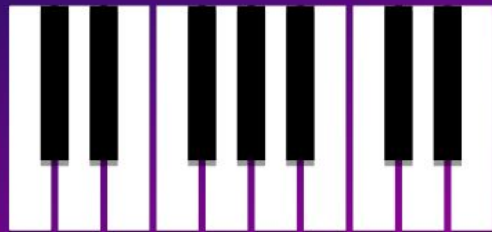
Intro to Rhythm Game Utilities plugin

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Rhythm Game Utilities

A collection of utilities for creating rhythm games in
Unity, Unreal, Godot, SDL and MonoGame.



<https://rhythmgameutilities.com/>

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Intro to song charting

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Intro to rhythm game data

Charting Songs

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Charting songs



notes.chart

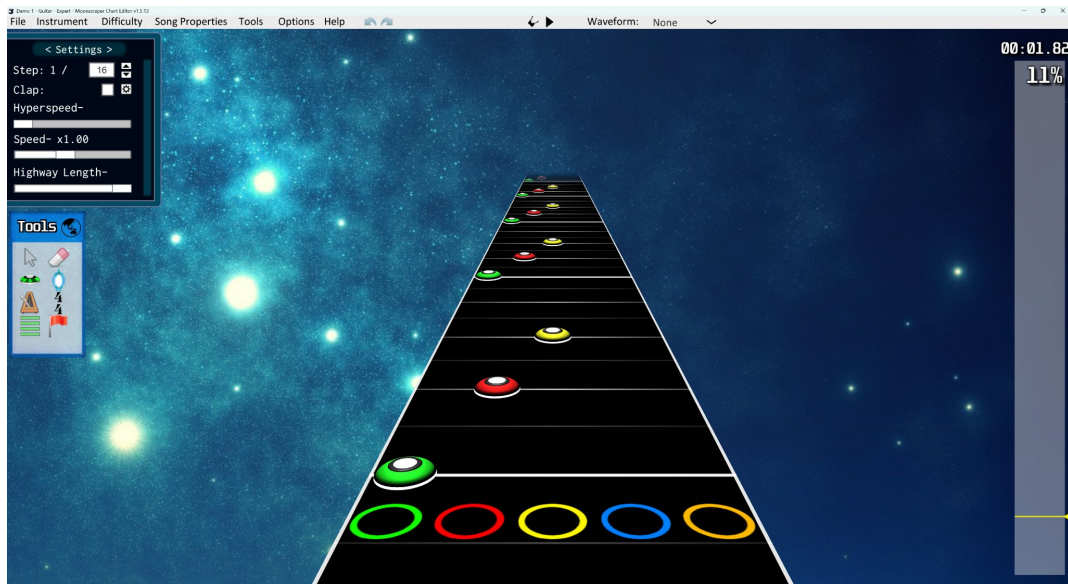


notes.mid

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Charting songs (.chart files)

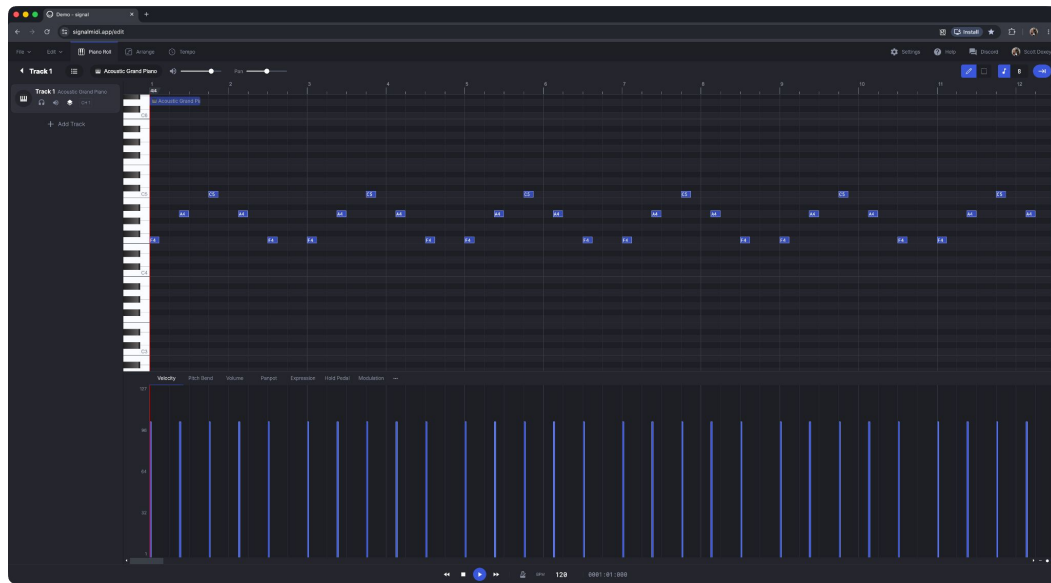


<https://github.com/FireFox2000000/Moonscraper-Chart-Editor>

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Charting songs (.mid files)



<https://signalmidi.app/>

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Intro to rhythm game data

Loading Songs

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Loading Notes (.chart)



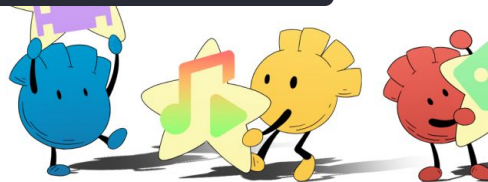
notes.chart



```
[Song]
{
  Name = "Demo 1"
  Artist = "Scott Doxey"
  Charter = "Scott Doxey"
  Album = "Demo"
  Year = ", 2024"
  Offset = 0
  Resolution = 192
  Player2 = bass
  Difficulty = 0
  PreviewStart = 0
  PreviewEnd = 0
  Genre = "rock"
  MediaType = "cd"
  MusicStream = "song.ogg"
}

[SyncTrack]
{
  0 = TS 4
  0 = B 120000
}

[ExpertSingle]
{
  768 = N 0 0
  960 = N 1 0
  1152 = N 2 0
  1536 = N 0 0
  1728 = N 1 0
  1920 = N 2 0
  2304 = N 0 0
  2496 = N 1 0
  2688 = N 2 0
  3072 = N 0 0
  3264 = N 1 0
  3456 = N 2 0
  3840 = N 0 0
  4032 = N 1 0
  4224 = N 2 0
  4608 = N 0 0
  4800 = N 1 0
  4992 = N 2 0
  5376 = N 0 0
  5568 = N 1 0
  5760 = N 2 0
}
```

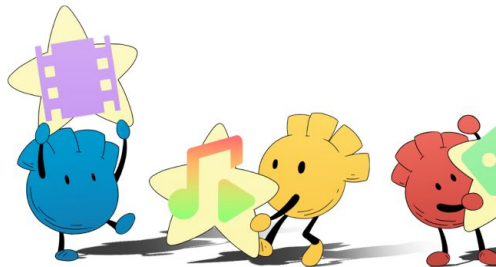


Loading Notes (.chart)

```
[SyncTrack]
{
  0 = TS 4
  0 = B 120000
}

[ExpertSingle]
{
  768 = N 0 0
  960 = N 1 0
  1152 = N 2 0
  1536 = N 0 0
  ...
}
```

First value is the
tick (position)



Loading Notes (.chart)

```
[SyncTrack]
{
  0 = TS 4           TS = Time Signature
  0 = B 120000       B = BPM
}
```



Loading Notes (.chart)

```
[ExpertSingle]
```

```
{
```

```
768 = N 0 0
```

```
960 = N 1 0
```

```
1152 = N 2 0
```

```
1536 = N 0 0
```

```
...
```

```
}
```

Note
Hand Position
Length (Hold Notes)

N 1 0



Loading Notes



notes.chart



notes.mid

```
{
  "Name": "Demo 1",
  "Artist": "Scott Doxey",
  "Album": "Demo",
  "Genre": "rock",
  "Year": " ", 2024,
  "Charter": "Scott Doxey",
  "Resolution": 192,
  "Difficulty": 0,
  "Offset": 0,
  "PreviewStart": 0,
  "PreviewEnd": 0,
  "MusicStream": "song.ogg",
  "Lyrics": {},
  ...
}

{
  ...,
  "BPM": {
    "0": 120000
  },
  "TimeSignatures": {
    "0": [
      4
    ]
  }
}

{
  ...,
  "Difficulties": {
    ...,
    "Expert": [
      {
        "Position": 768,
        "HandPosition": 0,
        "Length": 0
      },
      {
        "Position": 960,
        "HandPosition": 1,
        "Length": 0
      },
      {
        "Position": 1152,
        "HandPosition": 2,
        "Length": 0
      },
      {
        "Position": 1536,
        "HandPosition": 0,
        "Length": 0
      },
      ...
    ]
  },
  ...
}
```



Disclaimer:

The demo we are building is not optimized for performance or using best practices.

The code is optimized for this workshop to make understanding the concepts easier in the restricted time frame.

This demo is without juice.



Workshop

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>



Questions?

GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>





GodotCon Boston 2026 - <https://doxey.link/gcb-2026-rhythm>

